

Creation Date 24-Nov-2010

Revision Date 07-Dec-2024

Revision Number 5

Section 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description: Trimethylamine, 1M solution in THF
Cat No. : H32899

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Company

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address

begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99
CHEMTREC Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

Poison Centre - Emergency information services

Ireland : National Poisons Information Centre (NPIC) -
01 809 2166 (8am-10pm, 7 days a week)
Malta : +356 2395 2000
Cyprus : +357 2240 5611

Section 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

Physical hazards

Flammable liquids

Category 2 (H225)

Health hazards

Acute oral toxicity

Skin Corrosion/Irritation

Serious Eye Damage/Eye Irritation

Carcinogenicity

Specific target organ toxicity - (single exposure)

Category 4 (H302)

Category 1 B (H314)

Category 1 (H318)

Category 2 (H351)

Category 3 (H335) (H336)

Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Danger

Hazard Statements

H225 - Highly flammable liquid and vapor

H302 - Harmful if swallowed

H314 - Causes severe skin burns and eye damage

H335 - May cause respiratory irritation

H336 - May cause drowsiness or dizziness

H351 - Suspected of causing cancer

EUH019 - May form explosive peroxides

Precautionary Statements

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor/physician

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

2.3. Other hazards

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

Section 3: Composition/information on ingredients

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Tetrahydrofuran	109-99-9	203-726-8	93	Flam. Liq. 2 (H225) Acute Tox. 4 (H302) Eye Irrit. 2 (H319) STOT SE 3 (H335) STOT SE 3 (H336) Carc. 2 (H351) (EUH019)
Trimethylamine	75-50-3	EEC No. 200-875-0	7	Flam. Liq. 1 (H224) Skin Corr. 1B (H314) Eye Dam. 1 (H318) Acute Tox. 4 (H302) Acute Tox. 4 (H332)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Tetrahydrofuran	Acute Tox. 4 :: C>82.5% Eye Irrit. 2 :: C>=25% STOT SE 3 :: C>=25%	-	-
Trimethylamine	Eye Dam. 1 (H318) :: C>=5% Eye Irrit. 2 (H319) :: 0.5%<=C<5% Skin Irrit. 2 (H315) :: C>=5% STOT SE 3 (H335) :: C>=5%	-	-

Full text of Hazard Statements: see section 16

Section 4: First aid measures

4.1. Description of first aid measures

General Advice	Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.
Ingestion	Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.
Inhalation	If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.
Self-Protection of the First Aider	Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

Causes burns by all exposure routes. Difficulty in breathing. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Causes central nervous system depression

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically. Symptoms may be delayed.

Section 5: Firefighting measures

5.1. Extinguishing media

Suitable Extinguishing Media

Water mist may be used to cool closed containers. CO₂, dry chemical, dry sand, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

Do not use a solid water stream as it may scatter and spread fire.

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

Hazardous Combustion Products

Nitrogen oxides (NO_x), Carbon monoxide (CO), Carbon dioxide (CO₂), Thermal decomposition can lead to release of irritating gases and vapors.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

Section 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

6.3. Methods and material for containment and cleaning up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

Section 7: Handling and storage

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

7.1. Precautions for safe handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. If peroxide formation is suspected, do not open or move container. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Shelf life 12 months. May form explosive peroxides on prolonged storage. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Keep away from heat, sparks and flame. Flammables area.

Technical Rules for Hazardous Substances (TRGS) 510
Storage Class (LGK) (Germany)

Class 3

Switzerland - Storage of hazardous substances

Storage class - SC 3
<https://www.kvu.ch/de/themen/stoffe-und-produkte>
<https://www.kvu.ch/fr/themes/substances-et-produits>
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

7.3. Specific end use(s)

Use in laboratories

Section 8: Exposure controls/personal protection

8.1. Control parameters

Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
Tetrahydrofuran	TWA: 50 ppm (8h) TWA: 150 mg/m ³ (8h) STEL: 100 ppm (15min) STEL: 300 mg/m ³ (15min) Skin	STEL: 100 ppm 15 min STEL: 300 mg/m ³ 15 min TWA: 50 ppm 8 hr TWA: 150 mg/m ³ 8 hr Skin	TWA / VME: 50 ppm (8 heures). restrictive limit TWA / VME: 150 mg/m ³ (8 heures). restrictive limit STEL / VLCT: 100 ppm. restrictive limit STEL / VLCT: 300 mg/m ³ . restrictive limit Peau	TWA: 50 ppm 8 uren TWA: 150 mg/m ³ 8 uren STEL: 100 ppm 15 minuten STEL: 300 mg/m ³ 15 minuten Huid	STEL / VLA-EC: 100 ppm (15 minutos). STEL / VLA-EC: 300 mg/m ³ (15 minutos). TWA / VLA-ED: 50 ppm (8 horas) TWA / VLA-ED: 150 mg/m ³ (8 horas) Piel
Trimethylamine			TWA / VME: 2 ppm (8 heures). restrictive limit TWA / VME: 4.9 mg/m ³ (8 heures). restrictive limit STEL / VLCT: 5 ppm.	TWA: 2 ppm 8 uren TWA: 4.9 mg/m ³ 8 uren STEL: 5 ppm 15 minuten STEL: 12.5 mg/m ³ 15 minuten	STEL / VLA-EC: 5 ppm (15 minutos). STEL / VLA-EC: 12.5 mg/m ³ (15 minutos). TWA / VLA-ED: 2 ppm (8 horas)

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

			restrictive limit STEL / VLCT: 12.5 mg/m³. restrictive limit		TWA / VLA-ED: 4.9 mg/m³ (8 horas)
--	--	--	--	--	--------------------------------------

Component	Italy	Germany	Portugal	The Netherlands	Finland
Tetrahydrofuran	TWA: 50 ppm 8 ore. Time Weighted Average TWA: 150 mg/m³ 8 ore. Time Weighted Average STEL: 100 ppm 15 minuti. Short-term STEL: 300 mg/m³ 15 minuti. Short-term Pelle	TWA: 50 ppm (8 Stunden). AGW - exposure factor 2 TWA: 150 mg/m³ (8 Stunden). AGW - exposure factor 2 TWA: 20 ppm (8 Stunden). MAK TWA: 60 mg/m³ (8 Stunden). MAK Höhepunkt: 40 ppm Höhepunkt: 120 mg/m³ Haut	STEL: 100 ppm 15 minutos STEL: 300 mg/m³ 15 minutos TWA: 50 ppm 8 horas TWA: 150 mg/m³ 8 horas Pele	huid STEL: 200 ppm 15 minuten STEL: 600 mg/m³ 15 minuten TWA: 100 ppm 8 uren TWA: 300 mg/m³ 8 uren	TWA: 50 ppm 8 tunteina TWA: 150 mg/m³ 8 tunteina STEL: 100 ppm 15 minuutteina STEL: 300 mg/m³ 15 minuutteina Iho
Trimethylamine	TWA: 4.9 mg/m³ 8 ore. Time Weighted Average TWA: 2 ppm 8 ore. Time Weighted Average STEL: 12.5 mg/m³ 15 minuti. Short-term STEL: 5 ppm 15 minuti. Short-term	TWA: 2 ppm (8 Stunden). AGW - ceiling factor 2.5; exposure factor 2 TWA: 4.9 mg/m³ (8 Stunden). AGW - ceiling factor 2.5; exposure factor 2 TWA: 2 ppm (8 Stunden). MAK an instantaneous value of 5 ppm corresponding to 12 mg/m³ should not be exceeded;even if the MAK value is adhered to, "odor-associated" symptoms cannot be ruled out in individual cases TWA: 4.9 mg/m³ (8 Stunden). MAK an instantaneous value of 5 ppm corresponding to 12 mg/m³ should not be exceeded;even if the MAK value is adhered to, "odor-associated" symptoms cannot be ruled out in individual cases Höhepunkt: 4 ppm Höhepunkt: 9.8 mg/m³	STEL: 5 ppm 15 minutos STEL: 12.5 mg/m³ 15 minutos TWA: 2 ppm 8 horas TWA: 4.9 mg/m³ 8 horas	STEL: 5 ppm 15 minuten STEL: 12.5 mg/m³ 15 minuten TWA: 2 ppm 8 uren TWA: 4.9 mg/m³ 8 uren	TWA: 2 ppm 8 tunteina TWA: 4.9 mg/m³ 8 tunteina STEL: 5 ppm 15 minuutteina STEL: 12.5 mg/m³ 15 minuutteina

Component	Austria	Denmark	Switzerland	Poland	Norway
Tetrahydrofuran	Haut MAK-KZGW: 100 ppm 15 Minuten MAK-KZGW: 300 mg/m³ 15 Minuten MAK-TMW: 50 ppm 8 Stunden MAK-TMW: 150 mg/m³ 8 Stunden	TWA: 50 ppm 8 timer TWA: 150 mg/m³ 8 timer STEL: 300 mg/m³ 15 minutter STEL: 100 ppm 15 minutter Hud	Haut/Peau STEL: 100 ppm 15 Minuten STEL: 300 mg/m³ 15 Minuten TWA: 50 ppm 8 Stunden TWA: 150 mg/m³ 8 Stunden	STEL: 300 mg/m³ 15 minutach TWA: 150 mg/m³ 8 godzinach	TWA: 50 ppm 8 timer TWA: 150 mg/m³ 8 timer STEL: 75 ppm 15 minutter. value calculated STEL: 187.5 mg/m³ 15 minutter. value calculated Hud
Trimethylamine	MAK-KZGW: 5 ppm 15 Minuten MAK-KZGW: 12.5 mg/m³ 15 Minuten MAK-TMW: 2 ppm 8 Stunden MAK-TMW: 4.9 mg/m³ 8 Stunden	TWA: 2 ppm 8 timer TWA: 4.9 mg/m³ 8 timer STEL: 12.5 mg/m³ 15 minutter STEL: 5 ppm 15 minutter	STEL: 5 ppm 15 Minuten STEL: 12 mg/m³ 15 Minuten TWA: 2 ppm 8 Stunden TWA: 5 mg/m³ 8 Stunden	STEL: 12.5 mg/m³ 15 minutach TWA: 4.9 mg/m³ 8 godzinach	TWA: 4.9 mg/m³ 8 timer TWA: 2 ppm 8 timer STEL: 9.8 mg/m³ 15 minutter. value calculated STEL: 10 ppm 15 minutter. value calculated

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Tetrahydrofuran	TWA: 50.0 ppm TWA: 150.0 mg/m ³ STEL : 100 ppm STEL : 300.0 mg/m ³ Skin notation	kože TWA-GVI: 50 ppm 8 satima. TWA-GVI: 150 mg/m ³ 8 satima. STEL-KGVI: 100 ppm 15 minutama. STEL-KGVI: 300 mg/m ³ 15 minutama.	TWA: 50 ppm 8 hr. TWA: 150 mg/m ³ 8 hr. STEL: 100 ppm 15 min STEL: 300 mg/m ³ 15 min Skin	Skin-potential for cutaneous absorption STEL: 100 ppm STEL: 300 mg/m ³ TWA: 50 ppm TWA: 150 mg/m ³	TWA: 150 mg/m ³ 8 hodinách. Potential for cutaneous absorption Ceiling: 300 mg/m ³
Trimethylamine	TWA: 4.9 mg/m ³ TWA: 2 ppm STEL : 12.5 mg/m ³ STEL : 5 ppm	TWA-GVI: 2 ppm 8 satima. TWA-GVI: 4.9 mg/m ³ 8 satima. STEL-KGVI: 5 ppm 15 minutama. STEL-KGVI: 12.5 mg/m ³ 15 minutama.	TWA: 2 ppm 8 hr. TWA: 4.9 ppm 8 hr. STEL: 5 ppm 15 min STEL: 12.5 mg/m ³ 15 min	STEL: 12.5 mg/m ³ STEL: 5 ppm TWA: 4.9 mg/m ³ TWA: 2 ppm	TWA: 10 mg/m ³ 8 hodinách. Ceiling: 20 mg/m ³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Tetrahydrofuran	Nahk TWA: 50 ppm 8 tundides. TWA: 150 mg/m ³ 8 tundides. STEL: 100 ppm 15 minutites. STEL: 300 mg/m ³ 15 minutites.	Skin notation TWA: 50 ppm 8 hr TWA: 150 mg/m ³ 8 hr STEL: 100 ppm 15 min STEL: 300 mg/m ³ 15 min	STEL: 250 ppm STEL: 735 mg/m ³ TWA: 200 ppm TWA: 590 mg/m ³	STEL: 300 mg/m ³ 15 percekben. CK STEL: 100 ppm 15 percekben. CK TWA: 150 mg/m ³ 8 órában. AK TWA: 50 ppm 8 órában. AK lehetséges borón keresztül felszívódás	STEL: 100 ppm STEL: 300 mg/m ³ TWA: 50 ppm 8 klukkustundum. TWA: 150 mg/m ³ 8 klukkustundum. Skin notation
Trimethylamine	TWA: 2 ppm 8 tundides. TWA: 4.9 mg/m ³ 8 tundides. STEL: 12.5 mg/m ³ 15 minutites. STEL: 5 ppm 15 minutites.		STEL: 5 ppm STEL: 12.5 mg/m ³ TWA: 2 ppm TWA: 4.9 mg/m ³	STEL: 12.5 mg/m ³ 15 percekben. CK STEL: 5 ppm 15 percekben. CK TWA: 4.9 mg/m ³ 8 órában. AK TWA: 2 ppm 8 órában. AK	STEL: 5 ppm STEL: 12.5 mg/m ³ TWA: 2 ppm 8 klukkustundum. TWA: 4.9 mg/m ³ 8 klukkustundum.

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Tetrahydrofuran	skin - potential for cutaneous exposure STEL: 100 ppm STEL: 300 mg/m ³ TWA: 50 ppm TWA: 150 mg/m ³	TWA: 50 ppm IPRD TWA: 150 mg/m ³ IPRD Oda STEL: 100 ppm STEL: 300 mg/m ³	Possibility of significant uptake through the skin TWA: 50 ppm 8 Stunden TWA: 150 mg/m ³ 8 Stunden STEL: 100 ppm 15 Minuten STEL: 300 mg/m ³ 15 Minuten	possibility of significant uptake through the skin TWA: 50 ppm TWA: 150 mg/m ³ STEL: 100 ppm 15 minuti STEL: 300 mg/m ³ 15 minuti	Skin notation TWA: 50 ppm 8 ore TWA: 150 mg/m ³ 8 ore STEL: 100 ppm 15 minute STEL: 300 mg/m ³ 15 minute
Trimethylamine	STEL: 12.5 mg/m ³ STEL: 5 ppm TWA: 4.9 mg/m ³ TWA: 2 ppm	TWA: 4.9 mg/m ³ IPRD TWA: 2 ppm IPRD Oda STEL: 12.5 mg/m ³ STEL: 5 ppm	TWA: 4.9 mg/m ³ 8 Stunden TWA: 2 ppm 8 Stunden STEL: 12.5 mg/m ³ 15 Minuten STEL: 5 ppm 15 Minuten	TWA: 2 ppm TWA: 4.9 mg/m ³ STEL: 5 ppm 15 minuti STEL: 12.5 mg/m ³ 15 minuti	TWA: 4.9 mg/m ³ 8 ore TWA: 2 ppm 8 ore STEL: 12.5 mg/m ³ 15 minute STEL: 5 ppm 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Tetrahydrofuran	MAC: 100 mg/m ³	Ceiling: 300 mg/m ³ Potential for cutaneous absorption TWA: 50 ppm TWA: 150 mg/m ³	TWA: 50 ppm 8 urah TWA: 150 mg/m ³ 8 urah Koža STEL: 100 ppm 15 minutah STEL: 300 mg/m ³ 15 minutah	Binding STEL: 100 ppm 15 minuter Binding STEL: 300 mg/m ³ 15 minuter TLV: 50 ppm 8 timmar. NGV TLV: 150 mg/m ³ 8 timmar. NGV	Deri TWA: 50 ppm 8 saat TWA: 150 mg/m ³ 8 saat STEL: 100 ppm 15 dakika STEL: 300 mg/m ³ 15 dakika
Trimethylamine	Skin notation		TWA: 4.9 mg/m ³ 8 urah	Binding STEL: 5 ppm 15	

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

	MAC: 5 mg/m ³		TWA: 2 ppm 8 urah STEL: 5 ppm 15 minutah STEL: 12.5 mg/m ³ 15 minutah	minuter Binding STEL: 12.5 mg/m ³ 15 minuter TLV: 2 ppm 8 timmar. NGV TLV: 4.9 mg/m ³ 8 timmar. NGV	
--	--------------------------	--	--	---	--

Biological limit values

List source(s):

Component	European Union	United Kingdom	France	Spain	Germany
Tetrahydrofuran				Tetrahydrofuran: 2 mg/L urine end of shift	Tetrahydrofuran: 2 mg/L urine (end of shift)

Component	Gibraltar	Latvia	Slovak Republic	Luxembourg	Turkey
Tetrahydrofuran			Tetrahydrofuran: 2 mg/L urine end of exposure or work shift		

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Tetrahydrofuran 109-99-9 (93)				DNEL = 12.6mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Tetrahydrofuran 109-99-9 (93)	DNEL = 300mg/m ³	DNEL = 96mg/m ³	DNEL = 150mg/m ³	DNEL = 72.4mg/m ³

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Tetrahydrofuran 109-99-9 (93)	PNEC = 4.32mg/L	PNEC = 23.3mg/kg sediment dw	PNEC = 21.6mg/L	PNEC = 4.6mg/L	PNEC = 2.13mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Tetrahydrofuran 109-99-9 (93)	PNEC = 0.432mg/L	PNEC = 2.33mg/kg sediment dw		PNEC = 67mg/kg food	

8.2. Exposure controls

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

Engineering Measures

Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Goggles (European standard - EN 166)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Butyl rubber	See manufacturers	-	EN 374	(minimum requirement)
Nitrile rubber	recommendations			
Viton (R)				
Neoprene gloves				

Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

Recommended Filter type: low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387

Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Recommended half mask:- Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls

No information available.

Section 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance	Clear	
Odor	No information available	
Odor Threshold	No data available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	No information available	
Flammability (liquid)	Highly flammable	On basis of test data
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Flash Point	-7 °C / 19.4 °F	Method - No information available
Autoignition Temperature	No data available	

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

Decomposition Temperature	No data available	
pH	No information available	
Viscosity	No data available	
Water Solubility	Miscible	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
Tetrahydrofuran	0.45	
Trimethylamine	-2.9 - 1.89	
Vapor Pressure	No data available	
Density / Specific Gravity	0.850	
Bulk Density	Not applicable	Liquid
Vapor Density	No information available	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Explosive Properties Vapors may form explosive mixtures with air

Section 10: Stability and reactivity

10.1. Reactivity None known, based on information available

10.2. Chemical stability Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization Hazardous polymerization does not occur.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.

10.5. Incompatible materials Strong oxidizing agents.

10.6. Hazardous decomposition products Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO₂). Thermal decomposition can lead to release of irritating gases and vapors.

Section 11: Toxicological information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;
Oral Category 4
Dermal Based on available data, the classification criteria are not met
Inhalation Based on available data, the classification criteria are not met

Toxicology data for the components

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Tetrahydrofuran	1650 mg/kg (Rat)	> 2000 mg/kg (Rabbit)	180 mg/L (Rat) 1 h 53.9 mg/L (Rat) 4 h
Trimethylamine	LD50 = 1200 mg/kg (Rat)	LD50 > 5000 mg/kg (Rat)	LC50 > 5.9 mg/L (Rat) 4 h

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;
 Respiratory No data available
 Skin No data available

Component	Test method	Test species	Study result
Tetrahydrofuran 109-99-9 (93)	Local Lymph Node Assay OECD Test Guideline 429	mouse	non-sensitising

(e) germ cell mutagenicity; No data available

Component	Test method	Test species	Study result
Tetrahydrofuran 109-99-9 (93)	OECD Test Guideline 476 Gene cell mutation	in vivo Mammalian	negative
	OECD Test Guideline 473 Chromosomal aberration assay	in vitro Mammalian	negative

(f) carcinogenicity; Category 2
 Limited evidence of a carcinogenic effect The table below indicates whether each agency has listed any ingredient as a carcinogen

Component	EU	UK	Germany	IARC
Tetrahydrofuran				Group 2B

(g) reproductive toxicity; No data available

Component	Test method	Test species / Duration	Study result
Tetrahydrofuran 109-99-9 (93)	OECD Test Guideline 416	Rat 2 Generation	NOAEL = 3,000 ppm

(h) STOT-single exposure; Category 3
 Results / Target organs Respiratory system, Central nervous system (CNS).

(i) STOT-repeated exposure; No data available
 Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Causes central nervous system depression.

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

11.2. Information on other hazards

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

Section 12: Ecological information

12.1. Toxicity

Ecotoxicity effects

Do not empty into drains. .

Component	Freshwater Fish	Water Flea	Freshwater Algae
Tetrahydrofuran	2160 mg/l LC50 = 96 h Pimephales promelas Leuciscus idus: LC50: 2820 mg/L/48h	EC50 48 h 3485 mg/l EC50: >10000 mg/L/24h	
Trimethylamine		EC50: = 139 mg/L, 48h (Daphnia magna Straus)	EC50: = 74.2 mg/L, 96h (Desmodesmus subspicatus) EC50: = 98.8 mg/L, 72h (Desmodesmus subspicatus)

Component	Microtox	M-Factor
Trimethylamine	EC50 = 210 mg/L 17 h EC50 = 590 mg/L 2 h	

12.2. Persistence and degradability

Persistence

Persistence is unlikely.

12.3. Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Tetrahydrofuran	0.45	No data available
Trimethylamine	-2.9 - 1.89	No data available

12.4. Mobility in soil

The product is water soluble, and may spread in water systems . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

12.5. Results of PBT and vPvB assessment

No data available for assessment.

12.6. Endocrine disrupting properties

Endocrine Disruptor Information

Component	EU - Endocrine Disruptors Candidate List	EU - Endocrine Disruptors - Evaluated Substances
Tetrahydrofuran	Group III Chemical	

12.7. Other adverse effects

Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance
This product does not contain any known or suspected substance

Section 13: Disposal considerations

13.1. Waste treatment methods

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

Waste from Residues/Unused Products	Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.
Contaminated Packaging	Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.
European Waste Catalogue (EWC)	According to the European Waste Catalog, Waste Codes are not product specific, but application specific.
Other Information	Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.
Switzerland - Waste Ordinance	Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600 https://www.fedlex.admin.ch/eli/cc/2015/891/en

Section 14: Transport information

IMDG/IMO

14.1. UN number	UN2924
14.2. UN proper shipping name	Flammable liquid, corrosive, n.o.s.
Technical Shipping Name	Tetrahydrofuran, Trimethylamine
14.3. Transport hazard class(es)	3
Subsidiary Hazard Class	8
14.4. Packing group	II

ADR

14.1. UN number	UN2924
14.2. UN proper shipping name	Flammable liquid, corrosive, n.o.s.
Technical Shipping Name	Tetrahydrofuran, Trimethylamine
14.3. Transport hazard class(es)	3
Subsidiary Hazard Class	8
14.4. Packing group	II

IATA

14.1. UN number	UN2924
14.2. UN proper shipping name	Flammable liquid, corrosive, n.o.s.
Technical Shipping Name	Tetrahydrofuran, Trimethylamine
14.3. Transport hazard class(es)	3
Subsidiary Hazard Class	8
14.4. Packing group	II

14.5. Environmental hazards	No hazards identified
14.6. Special precautions for user	No special precautions required.
14.7. Maritime transport in bulk according to IMO instruments	Not applicable, packaged goods

Section 15: Regulatory information

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Tetrahydrofuran	109-99-9	203-726-8	-	-	X	X	KE-33454	X	X
Trimethylamine	75-50-3	200-875-0	-	-	X	X	KE-11508	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Tetrahydrofuran	109-99-9	X	ACTIVE	X	-	X	X	X
Trimethylamine	75-50-3	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Tetrahydrofuran	109-99-9	-	Use restricted. See entry 75. (see link for restriction details)	-
Trimethylamine	75-50-3	-	Use restricted. See entry 75. (see link for restriction details)	-

REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Tetrahydrofuran	109-99-9	Not applicable	Not applicable
Trimethylamine	75-50-3	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

National Regulations

ALFAAH32899

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification Water endangering class = 1 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Tetrahydrofuran	WGK1	
Trimethylamine	WGK1	

Component	France - INRS (Tables of occupational diseases)
Tetrahydrofuran	Tableaux des maladies professionnelles (TMP) - RG 84
Trimethylamine	Tableaux des maladies professionnelles (TMP) - RG 49, RG 49bis

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Tetrahydrofuran 109-99-9 (93)		Group I	

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

Section 16: Other information

Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed
H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage
H335 - May cause respiratory irritation
H336 - May cause drowsiness or dizziness
H351 - Suspected of causing cancer
EUH019 - May form explosive peroxides
H224 - Extremely flammable liquid and vapor
H225 - Highly flammable liquid and vapor
H319 - Causes serious eye irritation
H332 - Harmful if inhaled

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

SAFETY DATA SHEET

Trimethylamine, 1M solution in THF

Revision Date 07-Dec-2024

PBT - Persistent, Bioaccumulative, Toxic

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards On basis of test data

Health Hazards Calculation method

Environmental hazards Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

Prepared By Health, Safety and Environmental Department

Creation Date 24-Nov-2010

Revision Date 07-Dec-2024

Revision Summary Not applicable.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet